

BAULKHAM HILLS HIGH SCHOOL



MATHEMATICS EXTENSION 2 ASSESSMENT

December 2012

*Time allowed: 50 minutes
plus 5 minutes reading time*

BOS NUMBER : _____

TEACHER'S NAME: _____

QUESTION	MARK
1	/10
2	/10
3	/7
4	/11
TOTAL	/38
PERCENTAGE	



Extension 2 Mathematics

December 2012

Time: 50 minutes + 5 minutes reading time

DIRECTIONS

- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Use black or blue pen only (*not pencils*) to write your solutions.
- No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.
- Approved Maths aids and calculators may be used
- Marks are indicated next to each question

QUESTION 1 Start on the appropriate page in your booklet (10 marks)

a If $u = 1 - i$ and $v = -2 + 4i$ evaluate the following

i. $u^2 + v$

1

ii. $\text{Im}(u + \bar{v})$

2

iii. $\left| \frac{u+v+1}{u+v+i} \right|$

2

b Let z the complex number $z = \sqrt{3} + i$

Express in mod-arg form (with principal argument)

i. z

2

ii. iz

1

iii. z^9

2

QUESTION 2 Start on the appropriate page in your booklet (10 marks)

a Find both square roots of $-3 + 4i$

2

b Sketch the locus of z if $\arg(z - 1) = \arg(z - i)$

2

c i. Sketch $|z - 2i| \leq 1$

2

ii. Find the largest value of $\arg(z)$ when $|z - 2i| = 1$

1

d Sketch the locus of z if $\text{Re}\left(\frac{-i}{z}\right) = 2 \text{Im}\left(-\frac{i}{z}\right)$

3

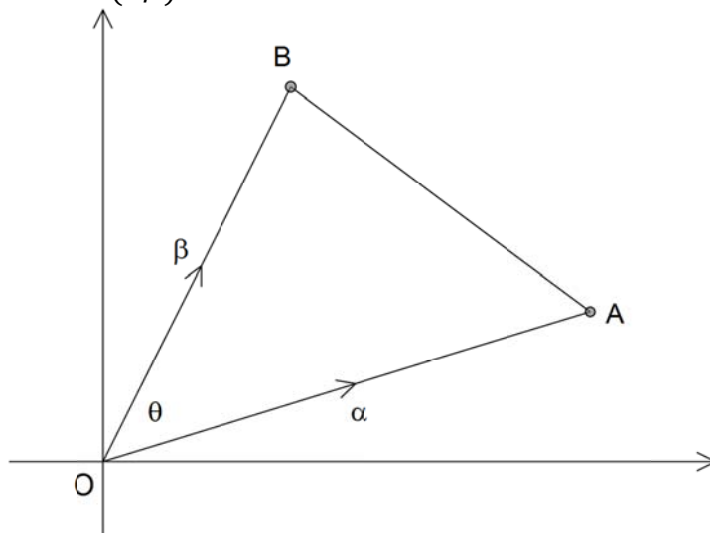
CONTINUED ON NEXT PAGE

QUESTION 3 Start on the appropriate page in your booklet (7 marks)

- a**
- i. Given $|z| = 1$, show that $z^n + z^{-n} = 2 \cos n\theta$. 2
 - ii. Solve $z^6 + 1 = 0$ and 3
hence factorise $z^6 + 1$ into real quadratic factors
 - iii. Hence deduce that $\cos 3\theta = \cos \theta (2 \cos \theta + \sqrt{3})(2 \cos \theta - \sqrt{3})$ 2

QUESTION 4 Start on the appropriate page in your booklet (11 marks)

- a** Given that $z + \frac{1}{z} = k$ where k is real and $z = x + iy$
- i. Show that either $y = 0$ or $x^2 + y^2 = 1$ 2
 - ii. Show that if $y = 0$ then $|k| \geq 2$ 2
 - iii. Show that if $x^2 + y^2 = 1$ then $|k| \leq 2$ 2
- b**
- i. Use the results $z + \bar{z} = 2\operatorname{Re}(z)$ and $|z|^2 = z\bar{z}$ to show that 3
 $| \alpha |^2 + | \beta |^2 - | \alpha - \beta |^2 = 2 \operatorname{Re}(\alpha \bar{\beta})$
 - ii. The diagram shows the angle θ between the complex numbers α and β 2
Prove that $| \alpha \beta | \cos \theta = \operatorname{Re}(\alpha \bar{\beta})$



~ END OF EXAM ~

Extension 2 DECEMBER TASK 2012

1 a) i) $(1-i)^2 - 2+4i$
 $= 1-2i+i^2-2+4i$
 $= -2+2i$ ✓

ii) $\text{Im}(1-i + (-2-4i))$ ① for conjugate

$= \text{Im}(-1-5i)$
 $= -5$ ✓

iii) $\frac{|1-i-2+4i+1|}{|1-i-2+4i+i|}$

$= \frac{|3i|}{|-1+4i|}$

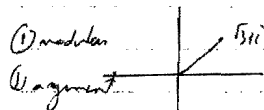
$= \frac{|3i|}{|-1+4i|}$

$= \frac{3}{\sqrt{1+4^2}}$

$= \frac{3}{\sqrt{17}}$ or $\frac{3\sqrt{17}}{17}$ ✓

or equivalent: $\frac{3i(-1-4i)}{(-1+4i)(-1-4i)}$ ✓

1 b) i) $z = |z|(\cos 30^\circ + i \sin 30^\circ)$
 $= \sqrt{1+1}(\cos 30^\circ + i \sin 30^\circ)$
 $z = 2(\cos 30^\circ + i \sin 30^\circ)$ ✓



ii) $iz = 2(\cos 120^\circ + i \sin 120^\circ)$ ✓

iii) $z^9 = (2(\cos 30^\circ + i \sin 30^\circ))^9$

$z = 512(\cos 270^\circ + i \sin 270^\circ)$ ✓

$z = 512(\cos(-90^\circ) + i \sin(-90^\circ))$ ✓

2 a) $x+iy = \sqrt{-3+4i}$

$x^2-y^2+2xyi = -3+4i$
 by inspection $\sqrt{-3+4i} = \pm(1+2i)$ ✓

(or $x^2-y^2 = -3$ $2xy = 4$
 $y = \frac{2}{x}$ ✓

$x^2 - \frac{4}{x^2} = -3$

$x^4 - 4 = -3x^2$

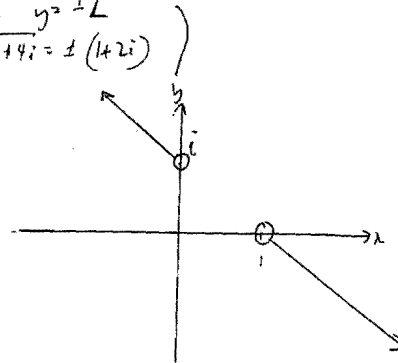
$x^4 + 3x^2 - 4 = 0$

$(x^2+4)(x^2-1) = 0$

no real soln $x^2-1=0$
 $x = \pm 1$

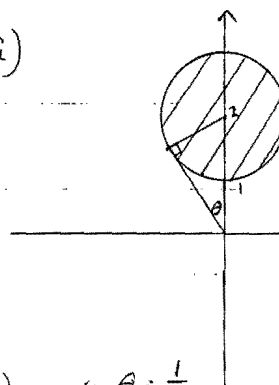
$\therefore y = \pm 2$
 $\therefore \sqrt{-3+4i} = \pm(1+2i)$ ✓

b)



- ① endpoints ✓
- ② line excluding interval from i to 1. ✓

c) i)



- ② circle, centre $2i$, with i or $3i$. ✓
- labelled, & shaded inside ✓

- ① progress toward (eg centre $2i$, shaded inside but no clear radius) ✓

ii) $\sin \theta = \frac{1}{2}$
 $\theta = 30^\circ$

$\therefore \text{max argument} = 120^\circ$ or $(\frac{2\pi}{3})$ ✓

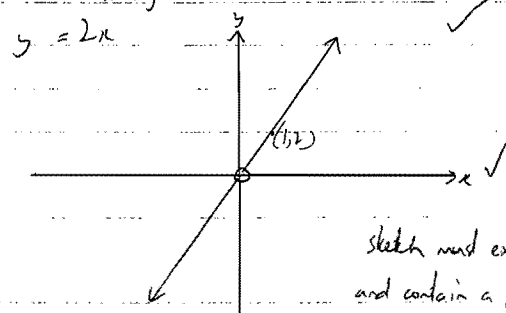
$$2d) \operatorname{Re}\left(\frac{-i}{z}\right) = 2 \operatorname{Im}\left(\frac{-i}{z}\right) \quad \text{NB } z \neq 0$$

$$\text{let } z = x + iy$$

$$\operatorname{Re}\left(\frac{-i(x-iy)}{z\bar{z}}\right) = 2 \operatorname{Im}\left(\frac{-i(x-iy)}{z\bar{z}}\right)$$

$$\operatorname{Re}\left(\frac{-y-ix}{x^2+y^2}\right) = 2 \operatorname{Im}\left(\frac{-y-ix}{x^2+y^2}\right)$$

$$\frac{-y}{x^2+y^2} = \frac{2(-x)}{x^2+y^2}$$



a) i) let $z = 1(\cos \theta + i \sin \theta)$ $|z|=1$
 $z^n + z^{-n} = (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n}$
 $= \cos(n\theta) + i \sin(n\theta) + \cos(-n\theta) + i \sin(-n\theta)$
 $= \cos(n\theta) + i \sin(n\theta) + \cos(n\theta) - i \sin(n\theta)$
 $z^n + z^{-n} = 2 \cos n\theta \quad \text{as req'd.}$

(ii) $z^6 = -1$
 $z_k = \operatorname{cis}\left(\frac{\pi + 2k\pi}{6}\right) \quad k=0,1,2,3,4,5 \quad (\text{or equivalent})$

$$z_0 = \operatorname{cis} \frac{\pi}{6}$$

$$z_1 = \operatorname{cis} \frac{3\pi}{6}$$

$$z_2 = \operatorname{cis} \frac{5\pi}{6}$$

$$z_3 = \operatorname{cis} \frac{7\pi}{6} = \bar{z}_2$$

$$z_4 = \operatorname{cis} \frac{9\pi}{6} = \bar{z}_1$$

$$z_5 = \operatorname{cis} \frac{11\pi}{6} = \bar{z}_0$$

$$k^2 - 4x + 1 \geq 0$$

$$k^2 - 4 \geq 0$$

$$k^2 \geq 4$$

$$|k| \geq 2$$

iii) $\left|z + \frac{1}{z}\right| = |k|$

$$\left|z + \frac{1}{z}\right| \geq |k|$$

$$\text{as } \left|z + \frac{1}{z}\right| \geq \left|z + \frac{1}{z}\right|$$

$$\left|z + \frac{1}{z}\right| \geq |k|$$

$$1 + \frac{1}{|z|} \geq |k|$$

$$\text{as } |z|=1 \text{ if } x^2+y^2=1$$

$$|k| \leq 2$$

(method 2: from above

$$(x+iy)(x^2+y^2) + x-iy = k(x^2+y^2)$$

$$x+iy + x-iy = k$$

$$2x = k$$

$$\text{Since } x^2+y^2=1 \quad -1 \leq x \leq 1$$

$$\therefore -2 \leq 2x \leq 2$$

$$-2 \leq k \leq 2$$

$$\text{ie } |k| \leq 2$$

b) i) $|a|^2 + |b|^2 = |a-b|^2$

$$= a\bar{a} + b\bar{b} = (a-b)(\overline{a-b})$$

$$= a\bar{a} + b\bar{b} = (a-b)(\bar{a}-\bar{b})$$

$$= a\bar{a} + b\bar{b} - (a\bar{b} + b\bar{a} - a\bar{b} - \bar{a}b)$$

$$= a\bar{b} + \bar{a}b$$

$$= a\bar{b} + \bar{a}b$$

$$= 2 \operatorname{Re}(a\bar{b})$$

$$z^4 + 1 = (z - z_1)(z - \bar{z}_1)(z - z_2)(z - \bar{z}_2)(z - z_3)(z - \bar{z}_3)$$

$$= (z^2 - (z_1 + \bar{z}_1)z + z_1 \bar{z}_1)(z^2 - (z_2 + \bar{z}_2)z + z_2 \bar{z}_2)(z^2 - (z_3 + \bar{z}_3)z + z_3 \bar{z}_3)$$

$$= (z^2 - 2 \cos \frac{\pi}{2} z + 1)(z^2 - 2 \cos \frac{2\pi}{2} z + 1)(z^2 - 2 \cos \frac{5\pi}{6} z + 1)$$

$$\text{ii)} \quad \frac{z^4 + 1}{z^3} = \frac{(z^2 + 1)}{z} \left(\frac{z^2 - \sqrt{3}z + 1}{z} \right) \left(\frac{z^2 + \sqrt{3}z + 1}{z} \right) \quad \text{OR}$$

$$z^3 + \frac{1}{z^3} = \left(z + \frac{1}{z}\right) \left(z + \frac{1}{z} - \sqrt{3}\right) \left(z + \frac{1}{z} + \sqrt{3}\right)$$

$$2 \cos 3\theta = 2 \cos \theta (2 \cos \theta - \sqrt{3}) (2 \cos \theta + \sqrt{3})$$

$$\cos 3\theta = \cos \theta (2 \cos \theta - \sqrt{3}) (2 \cos \theta + \sqrt{3})$$

$$\begin{aligned} (2 \cos \theta)^3 &= z^3 + \frac{1}{z^3} + 3 \left(z + \frac{1}{z}\right) \\ 8 \cos^3 \theta &= z^3 + \frac{1}{z^3} + 3 \left(z + \frac{1}{z}\right) \\ \cos 3\theta &= 4 \cos^3 \theta - 3 \cos \theta \\ \cos 3\theta &= \cos \theta (4 \cos^2 \theta - 3) \\ \cos 3\theta &= \cos \theta (2 \cos \theta + \sqrt{3}) (2 \cos \theta - \sqrt{3}) \end{aligned}$$

$$\text{Q4 a) i)} \quad z + \frac{1}{z} = k$$

$$x + iy + \frac{1}{x + iy} = k$$

$$x + iy + \frac{x - iy}{x^2 + y^2} = k$$

$$(x + iy) \frac{x^2 + y^2}{x^2 + y^2} + \frac{x - iy}{x^2 + y^2} = k \frac{(x^2 + y^2)}{x^2 + y^2} + 0i$$

$$\text{Equating imaginary parts } y(x^2 + y^2) - y = 0$$

$$y(x^2 + y^2 - 1) = 0$$

$$\therefore y = 0 \quad \text{or} \quad x^2 + y^2 - 1 = 0$$

$$\text{ie } x^2 + y^2 = 1$$

$$\text{ii)} \quad \text{If } y = 0 \quad z + \frac{1}{z} = k \text{ becomes}$$

$$x + \frac{1}{x} = k$$

$$\text{since } x \text{ and } k \text{ are real } \Delta \geq 0$$

$$x^2 + 1 = kx$$

$$x^2 - kx + 1 = 0$$

$$\text{b ii)} \quad \text{In } \triangle OBA$$

$$\cos \theta = \frac{|a|^2 + |b|^2 - |a - b|^2}{2|a||b|}$$

$$|a||b| \cos \theta = \frac{2 \operatorname{Re}(a\bar{b})}{2}$$

$$|a||b| \cos \theta = \operatorname{Re}(a\bar{b})$$